COMPUTER SYSTEMS AND PLATFORM TECHNOLOGIES

IT0047

MODULE 8

Operating System

SUBTOPIC 1

Introduction to Operating System

OBJECTIVES

Upon completion of this module, the student will be able to:

- Define the operating system (OS), its goals and services
- Demonstrate the abstract view of the components of a computer system;
- Enumerate the types of OS and the services provided by the operating systems
- Understand Basic Linux Commands, LINUX file system/structure, partition, filename format, absolute and relative paths
- Perform directory creation
- Display directory content using wildcards in LINUX commands
- Perform copy, rename and move files in linux
- Create text files using cat command and vi editor in linux
- Utilize different file operation, data refinement commands, and redirectors in file manipulation.
- Perform other Linux commands

WHAT IS AN OPERATING SYSTEM?

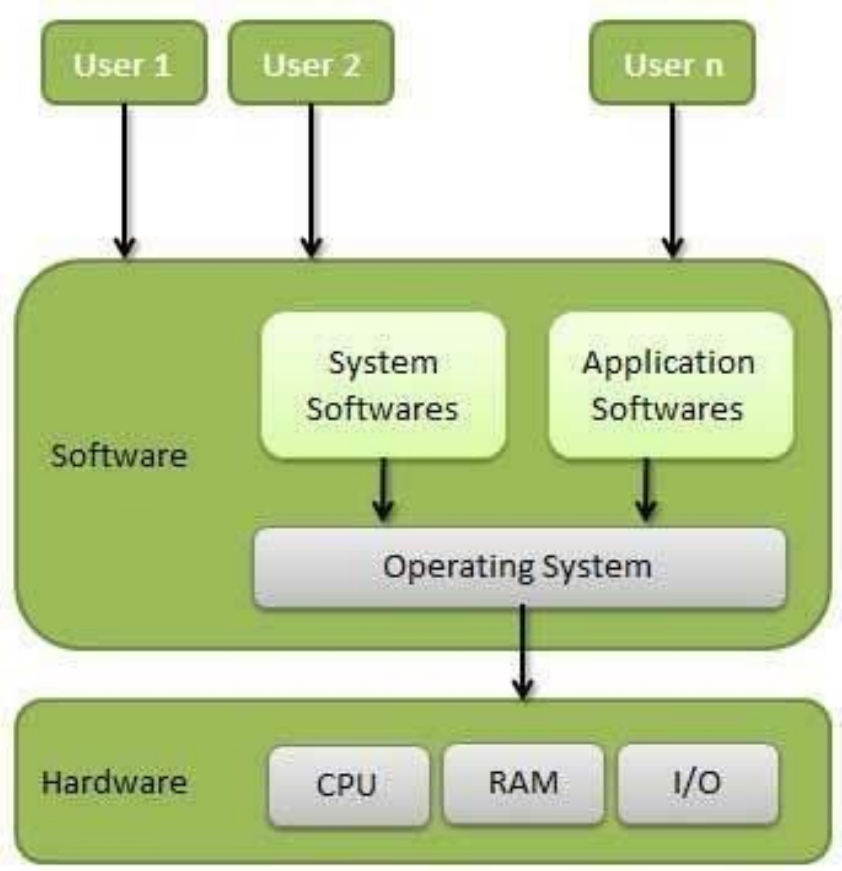
- An **Operating System (OS)** is a program or system software that acts as an interface between the user and the computer hardware and controls the execution of all kinds of programs.
- It is not possible for the user to use any computer or mobile device without having an operating system.

GOALS OF OPERATING SYSTEM

1. Execute user programs and make solving user problems easier.
2. Make the computer system convenient to use.
3. Use the computer hardware in an efficient manner.

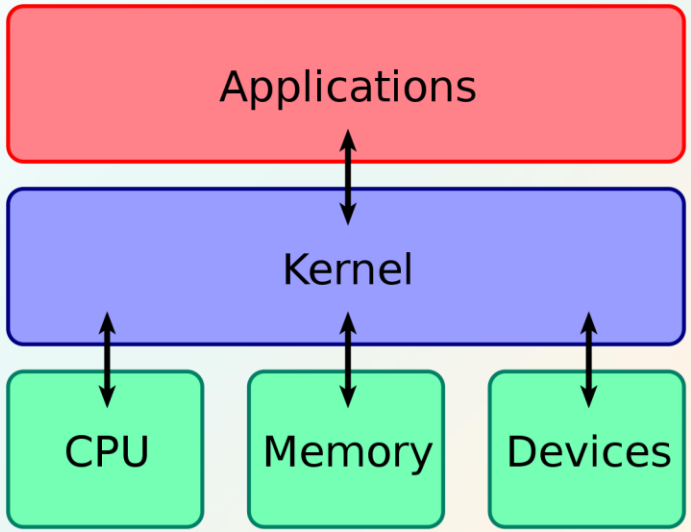
COMPONENTS OF A COMPUTER SYSTEM

- ***Computer hardware*** – CPU, memory and I/O devices, provides the basic computing resources.
- ***Application programs*** – are used to solve the computing problems of the users such as word processors, games and business programs .
- ***Users*** – who utilize a computer or network service trying to solve different problems.
- ***Operating System*** – controls and coordinates the use of the hardware among the various application programs for the various users.



WHAT IS A KERNEL?

Kernel is the central part of an OS which manages system resources and is always resident in memory. It also acts like a bridge between application and hardware of the computer. It is also the first program that loads after the **bootloader**.



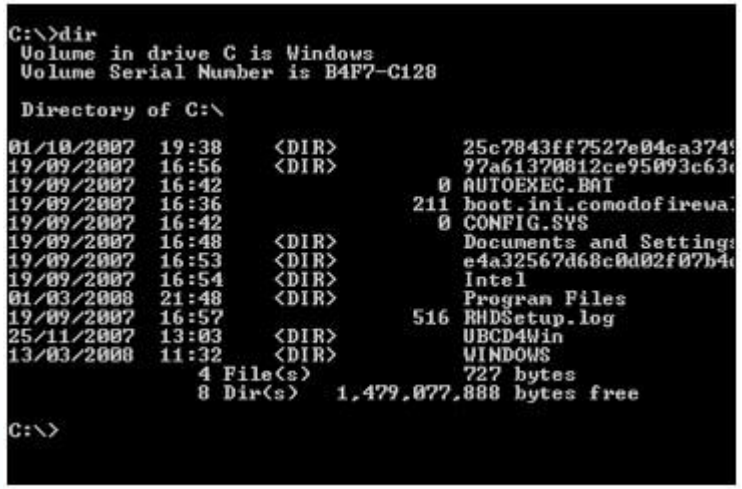
Bootloader is a program that loads and starts the boot time tasks and processes of an OS. It also places the OS of a computer into memory.

COMMON SERVICES OFFERED BY ALMOST ALL OPERATING SYSTEMS:

1. User Interface
2. Program Execution
3. File system manipulation
4. Input / Output Operations
5. Communication
6. Resource Allocation
7. Error Detection
8. Job Accounting
9. Security and protection

COMMON SERVICES OFFERED BY ALMOST ALL OPERATING SYSTEMS

1. **User Interface (UI).** Refers to the part of an OS, or device that allows user to enter and receive information.



CLI

GUI

Types of UI:

- Command line interface (CLI)
- Graphical User Interface (GUI)

COMMON SERVICES OFFERED BY ALMOST ALL OPERATING SYSTEMS:

2. **Program Execution.** The OS must have the capability to load a program into memory and execute that program.
3. **File system manipulation.** Program requires to read a file and write a file. File handling portion of OS allows users to create and delete files and directories.

COMMON SERVICES OFFERED BY ALMOST ALL OPERATING SYSTEMS:

4. **Input / Output Operations.** A program which is currently executing may require I/O, which may involve file or other I/O device. The OS is responsible for reading and/or writing data from I/O devices such as disks, tapes, printers, keyboards, etc.
5. **Communication.** Process needs to swap over information with other process. Processes executing on same computer system or on different computer systems can communicate using operating system support.

COMMON SERVICES OFFERED BY ALMOST ALL OPERATING SYSTEMS:

6. **Resource Allocation** The OS manages the different computer resources such as CPU time, memory space, file storage space, I/O devices, etc. and allocates them to different application programs and users.
7. **Error Detection** The operating system should be able to detect errors within the computer system (CPU, memory, I/O, or user program) and take the appropriate action.

COMMON SERVICES OFFERED BY ALMOST ALL OPERATING SYSTEMS:

8. **Job Accounting.** OS keeps track of time and resources used by various tasks and users, this information can be used to track resource usage for a particular user or group of user.
9. **Security and Protection.** *Protection* is any mechanism for controlling access of processes or users to resources defined by the OS. *Security* is a defense of the system against internal and external attacks (denial-of-service, worms, viruses, identity theft, theft of service)

TYPES OF OPERATING SYSTEM

Operating systems are there from the very first computer generation and they keep evolving with time.

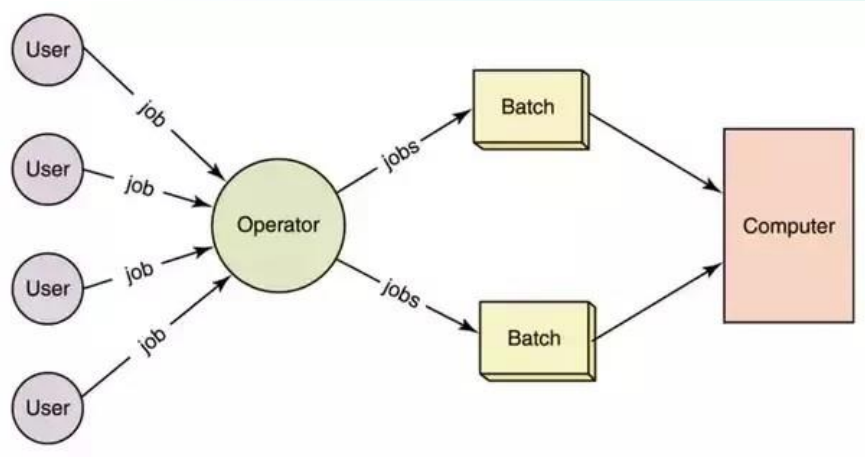
Some of the widely used operating systems are as follows:

1. Batch operating system
2. Time-sharing operating systems
3. Distributed operating system
4. Network operating system
5. Real-time operating system
6. Handheld operating system

TYPES OF OPERATING SYSTEM

1. Batch Operating System (BOS)

- The user of a BOS never directly interacts with the computer.
- Every user prepares his or her job on an offline device like a punch card and submit it to the computer operator.



- To speed up processing, jobs with similar needs are batched together and run as a group.
- The programmers leave their programs with the operator then operator sorts the programs with similar requirements into batches.

TYPES OF OPERATING SYSTEM

2. Time-sharing Operating System

- Time-sharing or **multitasking** is logical extension in which CPU switches jobs so frequently that users can interact with each job while it is running, creating **interactive computing.**
- Multiple jobs are executed by the CPU by switching between them, but the switches occur so frequently. Thus, the user can receive an immediate response. **Response time** should be < 1 second
- Each user has at least one program executing in memory
- If several jobs ready to run at the same time **CPU scheduling**
- If processes don't fit in memory, **swapping** will take place
- **Examples:** Unix, Linux, Multics and Windows

TYPES OF OPERATING SYSTEM

3. Distributed Operating System

- Distributed systems use multiple central processors to serve multiple real-time applications and multiple users. Data processing jobs are distributed among the processors accordingly.
- The processors communicate with one another through various communication lines (such as high-speed buses or telephone lines). These are referred as **loosely coupled systems** or **distributed systems**. Processors in a distributed system may vary in size and function. These processors are referred as sites, nodes, computers, and so on.
- **Examples:** Telecom Network, WWW, Cloud Computing, etc.

TYPES OF OPERATING SYSTEM

4. Network Operating System (NOS)

- A NOS runs on a server and provides the server the capability to manage data, users, groups, security, applications, and other networking functions.
- The primary purpose of the network operating system is to allow shared file and printer access among multiple computers in a network, typically a local area network (LAN), a private network or to other networks.
- **Examples:** Microsoft Windows Server 2003, Microsoft Windows Server 2008, UNIX, Linux, Mac OS X Server, Novell NetWare, and BSD/OS (Berkeley Software Design).

TYPES OF OPERATING SYSTEM

5. Real-time Operating System (RTOS)

- **RTOS** is an operating system intended to serve **real-time systems/applications** that process data as it comes in, mostly without buffer delay.
- It is a software component that rapidly switches between tasks, giving the impression that multiple programs are being executed at the same time on a single processing core.
- **Real-time systems** are used when there are time requirements like missile systems, air traffic control systems, robots, etc.
- **Examples:** LynxOS, OSE, QNX, RTLinux, VxWorks, Windows CE

TYPES OF OPERATING SYSTEM

6. Handheld Operating System

- It is also known as **Mobile OS** which is built exclusively for a mobile device, such as a smartphone, personal digital assistant (PDA), tablet, wearable devices or other embedded mobile OS.
- **Examples:** Android, Symbian, iOS, BlackBerry OS and Windows Mobile.

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Q&A SESSION

ASK ANY QUESTION RELATED TO OUR TOPIC FOR TODAY.

THANK YOU